



Sensor Fusion SLAM on the SubSurfaceGeoRobo Underground Datasets

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ROB 530/EECS 568: Mobile Robotics

Motivation

Underground mapping is essential for safe exploration and inspection of hazardous environments such as mines and tunnels, where GPS and wireless infrastructure are unavailable, but has many challenges.

- **Geometric Degeneracy**

Long straight tunnels exhibit a lack of longitudinal constraints, causing LiDAR drift along the heading direction [1]

- **Perceptual Degradation**

Airborne particulates (dust, smoke) and self-similar geometry degrade sensor fidelity. [1]

- **Environment Extremes**

High humidity and illumination changes visual or optical sensing systems [1]

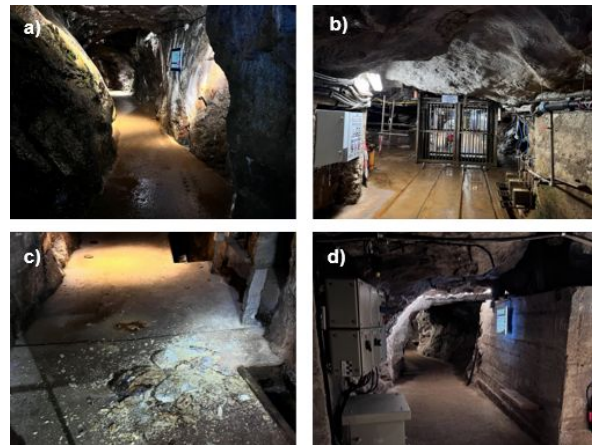
The Solution: Radar provides robustness in perceptually degraded underground environments



ARIDuA project underground mining robot

SubSurfaceGeoRobo Underground Dataset

- Published on October 15th, 2025 [2]
- 80 m ring-loop trajectory in the Reiche Zeche Research Mine
- Approximately 150 m underground
- Sensors
 - RS-LiDAR-16
 - Two Indurad ISDR-C mmWave radars
 - wheel odometry
 - IMU
- Ground truth: RIEGL VZ-400i TLS (< 2 mm accuracy)
- Narrowest passage (0.95 m) in published underground SLAM datasets



SubSurfaceGeoRobo Images

Technical Approach

1 LiDAR Preprocessing

- Range filter: 1–50 m (removes self-returns and far noise)
- Voxel downsample at 0.15 m
- Surface normal estimation for point-to-plane ICP

2 Local Submap + Scan Matching

- Sliding 20-scan submap
- Point-to-plane ICP at 1.0 m; if fitness ≤ 0.3 , retry at 3.0 m + point-to-point at 2.0 m, pick best

3 Degradation Simulation

- Standard: 8% dropout, 2% spurious, 0.02 m range noise
- Heavy: 25% dropout, 8% spurious, 0.05 m range noise
- Spurious points at random azimuths
- ICP runs on degraded scan; map built from clean scan to compare with ground truth

4 Evaluation

- Ground truth: RIEGL VZ-400i TLS point cloud (< 2 mm accuracy)
- Metric: cloud-to-cloud distance from SLAM map to GT
- Reported: Mean, Median, RMSE, 90th-percentile threshold

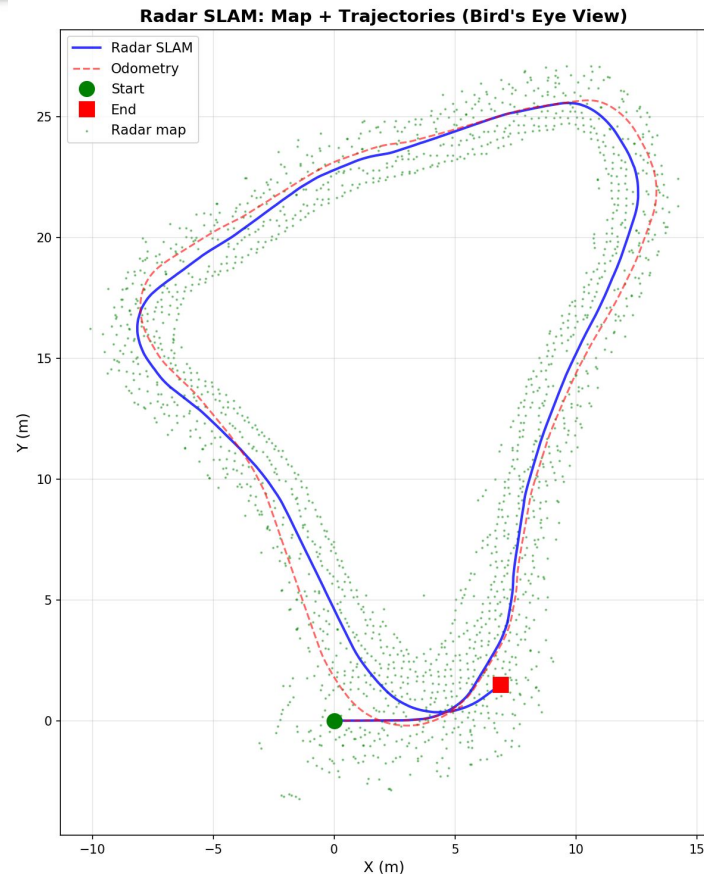
Approach: Radar Fusion

Radar-Only SLAM

- Accumulate radar hits within ± 4 s of scan, motion-compensate using wheel odometry, range-filter 0.5–5.0 m
- Radar submap: 10-scan sliding window at 0.15 m voxels; scan-to-submap ICP (point-to-point, coarse→fine)
- Odometry still provides motion prior; radar corrects with adaptive weight $w = \min(0.5, 0.6 \cdot \text{fitness})$

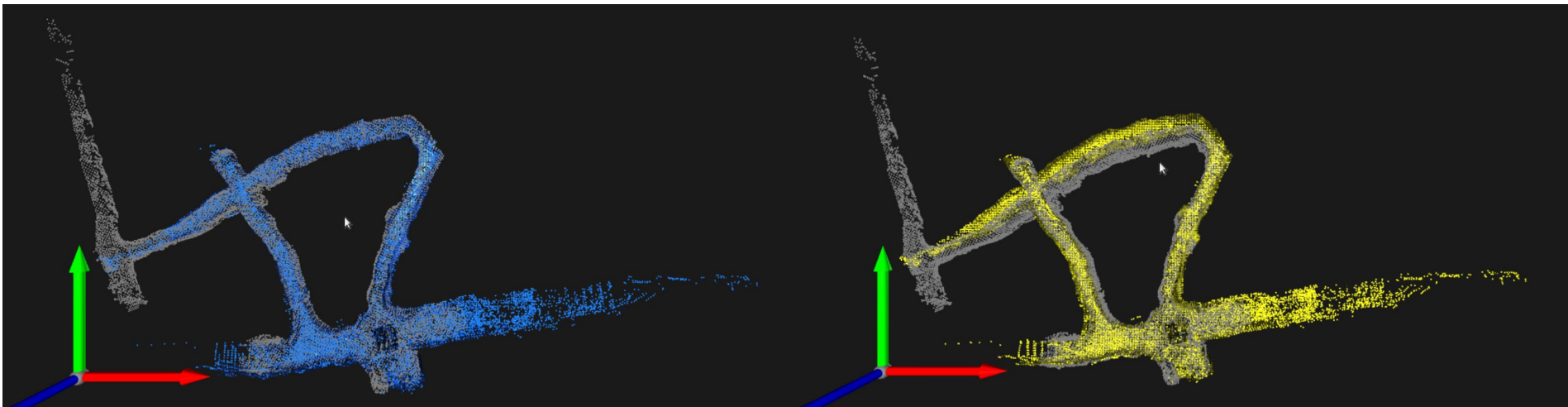
Fusion Method

- Run LiDAR ICP and radar ICP independently on their own submaps
- Blend pose estimates: translation = $(1-w) \cdot T_{\text{lidar}} + w \cdot T_{\text{radar}}$; yaw blended via shortest-arc delta
- Fusion weights determined by degradation level: 30% (clean), 70% (standard), 90% (heavy)



Radar-Only SLAM output: estimated trajectory (blue), wheel odometry prior (red), and radar point map (green).

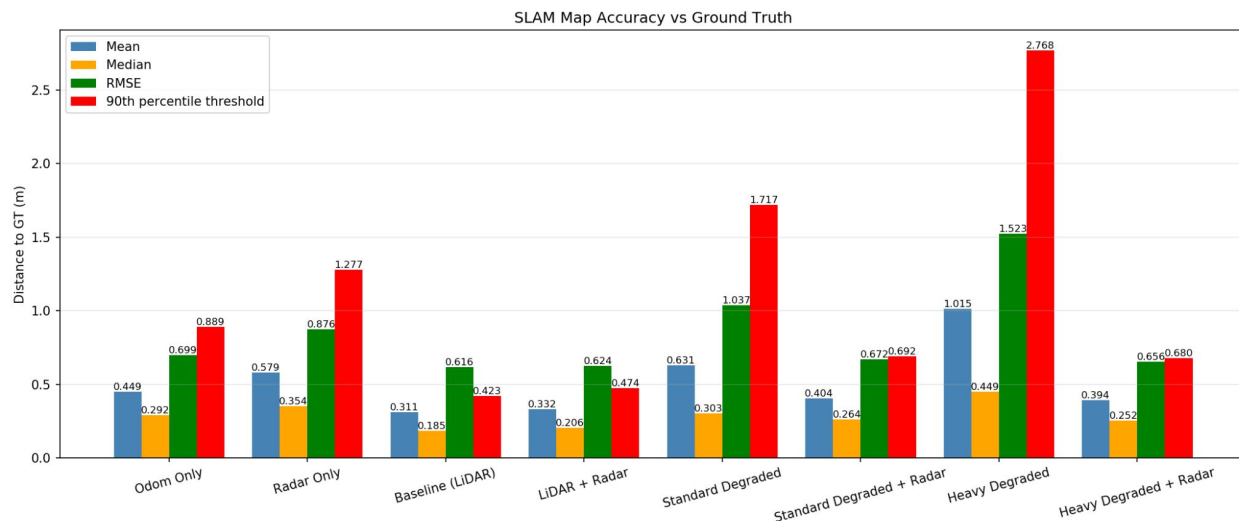
Results: 3D Map Visualization



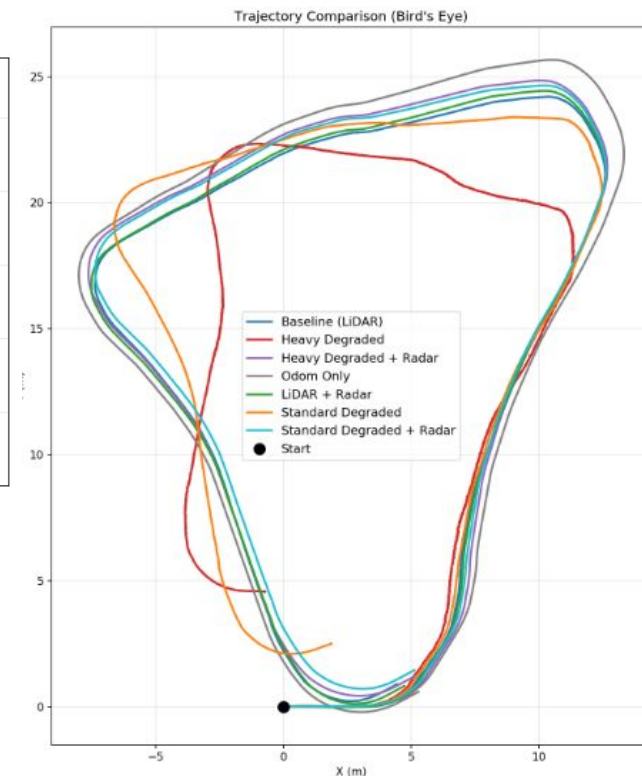
3D map comparison between baseline LiDAR SLAM (blue) and ground truth (grey)

3D map comparison between Odom only mapping (yellow) and ground truth (grey)

Results: Accuracy & Trajectory Comparison



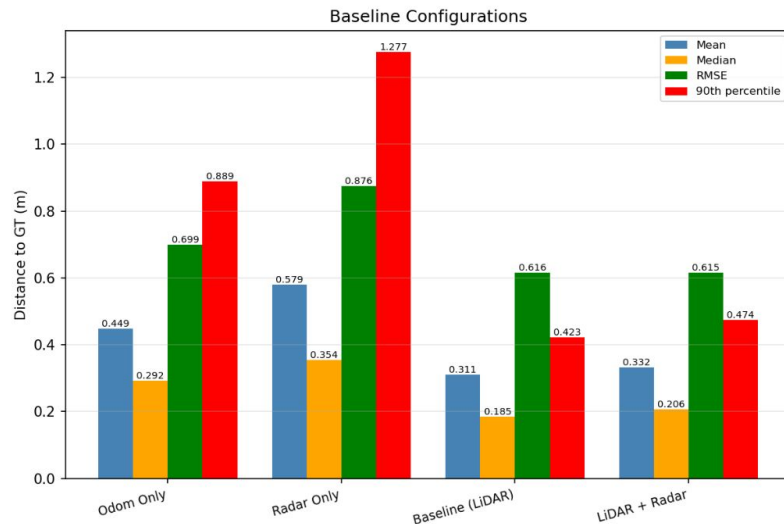
Slam Map Accuracies vs. Ground Truth



Trajectory Comparison

Analysis: Slam Map Accuracy

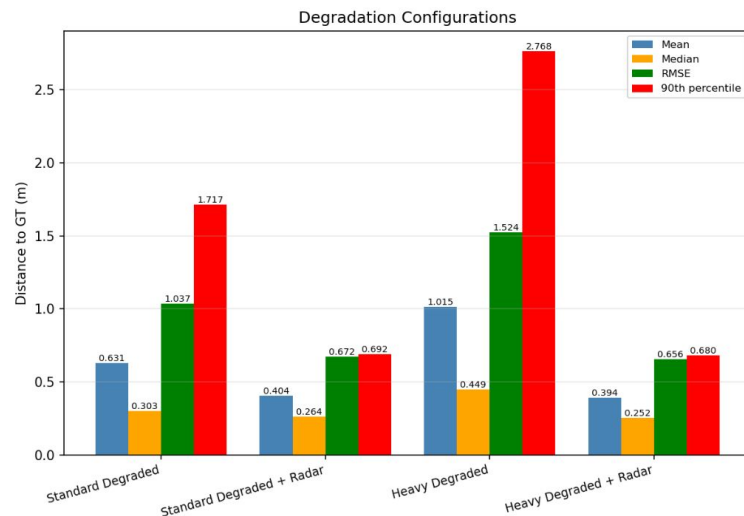
- **Baseline (0.311 m)** achieves the highest accuracy, serving as the performance upper bound
- **Radar Only (0.579 m)** is good but limited by sparse 2-D sensor geometry
- **LiDAR + Radar (0.316 m)** matches baseline
 - Radar adds no overhead under clean conditions
- **Odometry Only (0.449 m)** underperforms all LiDAR-based methods
 - Emphasizes necessity of scan-matching corrections



Non-degraded Slam Map
Accuracies vs. Ground Truth

Analysis: Degradation

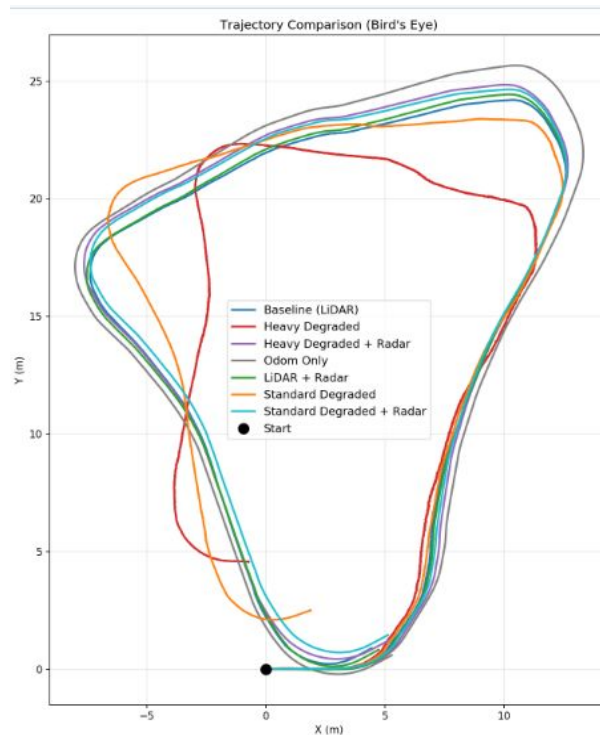
- **Standard Degraded (0.631 m)** significantly hurts accuracy and fails to close the loop
- **Heavy degradation** leads to widespread localization failure, with 90th percentile error reaching 2.768 m
- **Both recover substantially with radar fusion**
 - Heavy degradation sees bigger gains



Degraded Slam Map Accuracies vs.
Ground Truth

Analysis: Trajectory

- **Baseline, LiDAR+Radar, Heavy Degraded+Radar, and Standard Degraded+Radar** trajectories closely overlap, maintaining consistent path shape and clean loop closure
- **Radar fusion** effectively pulls degraded trajectories back to baseline-level paths, with recovery clearly visible in the bird's eye view



Conclusion

- LiDAR-only SLAM achieves the highest localization accuracy under normal conditions.
- Sensor degradation significantly hurts LiDAR-based localization
 - Severe degradation causing irreversible trajectory deformation.
- Radar fusion effectively recovers for lost LiDAR performance under degradation, restoring accuracy to near-baseline levels.
- Under non-degraded conditions, radar fusion introduces no measurable overhead, leaving system performance unaffected.
- The benefits of radar fusion increases with degradation severity
 - Adaptive weight adjustment is particularly critical under heavy degradation scenarios

Future Work

- Add loop closure to close the accuracy gap with state-of-the-art methods
- Replace 2-D Indurad radars with 4-D mmWave sensor to recover elevation constraints
- Test on additional underground sequences to validate generalization



Thank You

References

- [1] K. Ebadi et al. "Present and future of SLAM in extreme underground environments: The DARPA SubT Challenge," IEEE Transactions on Robotics, vol. 40, pp. 936–959, 2024, doi: 10.1109/TRO.2023.3323938.
- [2] J. Li et al., "SubSurfaceGeoRobo: A Comprehensive Underground Dataset for SLAM-based Geomonitoring with Sensor Calibration," PANGAEA, 2025. [Online]. Available: <https://doi.org/10.1594/PANGAEA.975532>